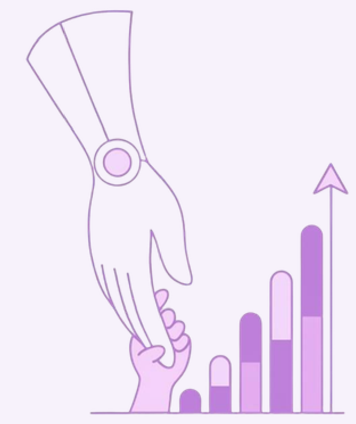
Princess Nourah bint Abdulrahman University
College of Computer and Information Sciences

HAYATY / حياتي

mHealth System for Early Detection of Child Developmental Delays

Princess Nourah bint Abdulrahman University | College of Computer and Information Sciences | GP2 · 2025/2026

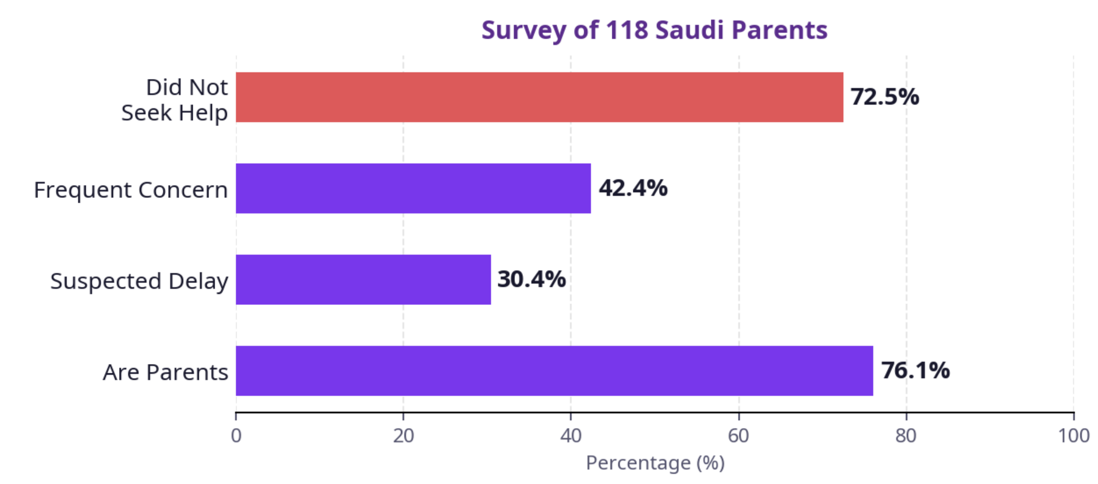
Motor | Cognitive | Social/Emotional | Language



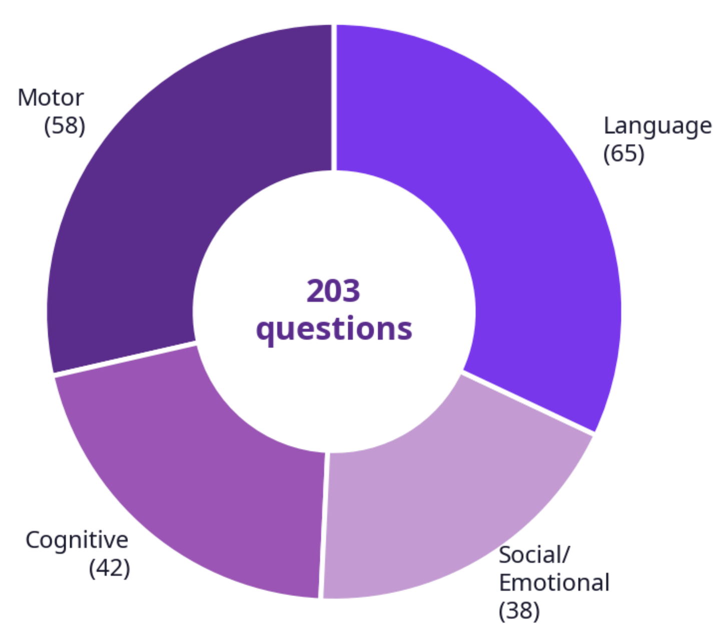
1 Abstract

HAYATY is an Arabic mHealth application for early detection of developmental delays in children aged 0–5 in Saudi Arabia. Parents complete age-based milestone checklists across four domains. AI speech analysis evaluates language milestones automatically. Computer vision detects gross motor milestones. Rule-based scoring classifies each domain as Natural, Follow-Up, or Refer. PDF progress reports are generated for healthcare professionals.

2 Introduction



72.5% who suspected a developmental delay did not seek professional evaluation. Language delays were the most frequently identified concern. Over 1,348 child apps exist yet none offer structured developmental assessment, Arabic support, or healthcare linkage. HAYATY fills this gap — transforming unstructured parental observations into structured, clinically shareable data.



Evidence Base

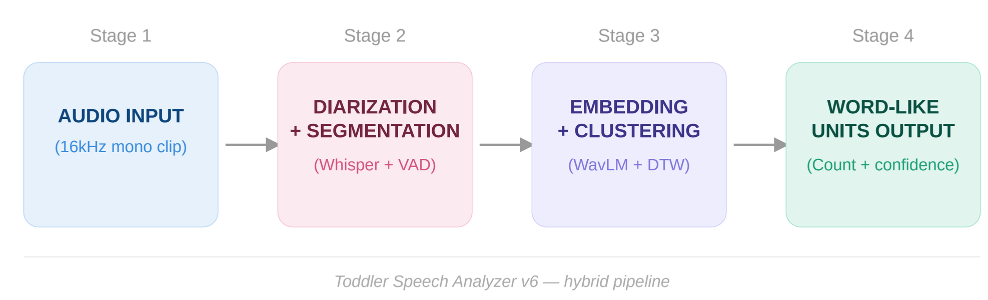
CDC Milestones 2022 | Bayley Scales III | WHO | MacArthur-Bates CDI | Denver II | ASQ-3

3 Methodology

Hybrid approach: rule-based evaluation + AI-powered analysis for speech and motor items.

Speech AI — Hybrid Child Speech Pipeline

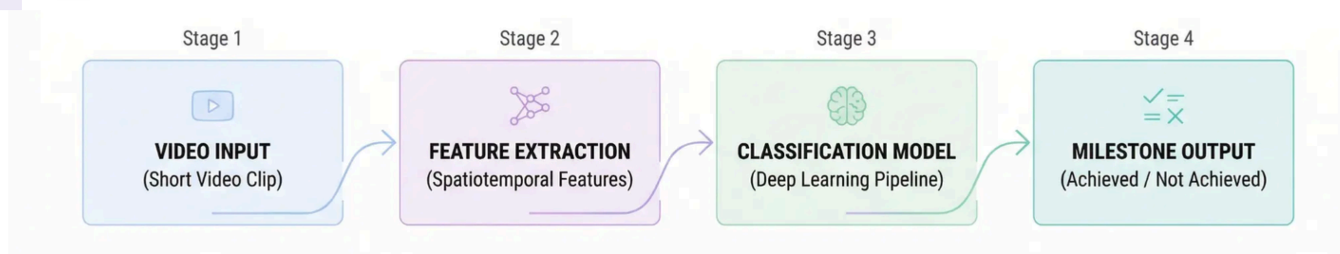
- Separates child speech from adult and overlap segments
- Uses Whisper-assisted segmentation to extract child utterances
- Combines WavLM embeddings with DTW-based clustering to discover repeated speech units
- Supports milestone-related analysis through high-confidence units



Motor CV — Rule-Based + Deep Learning

Phase 1 — MediaPipe Pose: geometric rules (postural, transition-based, locomotion). Result: 4/43 milestones stable, revealing fundamental limits of rule-based geometry for infant motion.

Phase 2 — Deep Learning: video clips → uniform frame sampling → pretrained backbone → 5-Fold Stratified CV → macro F1-score. Weighted loss + data augmentation for class imbalance.



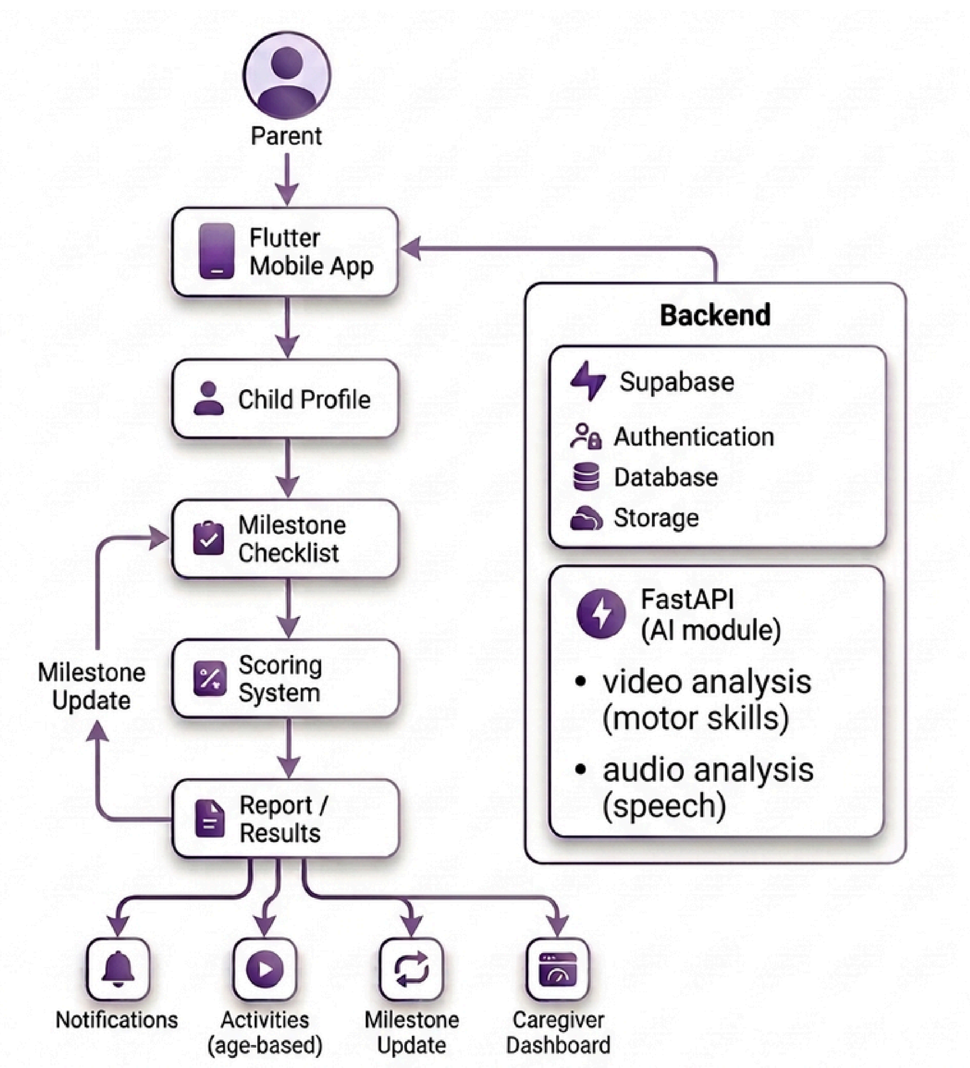
Response	Score	Classification	Action
Yes	0	Natural	No action
Sometimes	1	Follow-Up (2–3)	Re-assess 4–8 wks
No	2	Refer (≥4)	Immediate specialist

Notifications are triggered based on the child's total score:

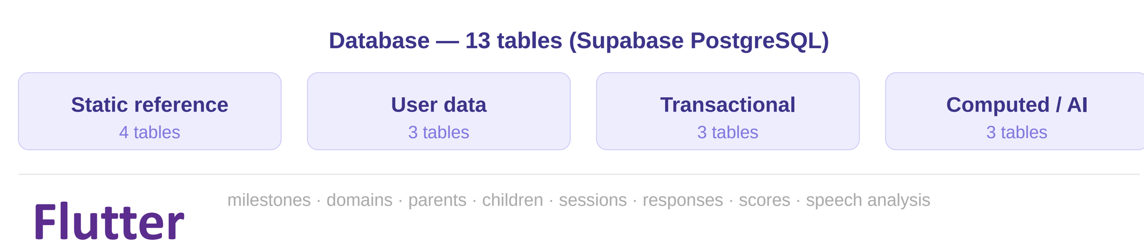
- Low score → Normal development
- Moderate score → Follow-up notification
- High score → Alert for potential delay and recommendation for specialist consultation

4 System Design

Three-tier: Flutter (Android+iOS, Arabic RTL) → Supabase (PostgreSQL 13 tables, Edge Functions) → FastAPI on Render. Firebase FCM alerts. Power BI caregiver analytics.

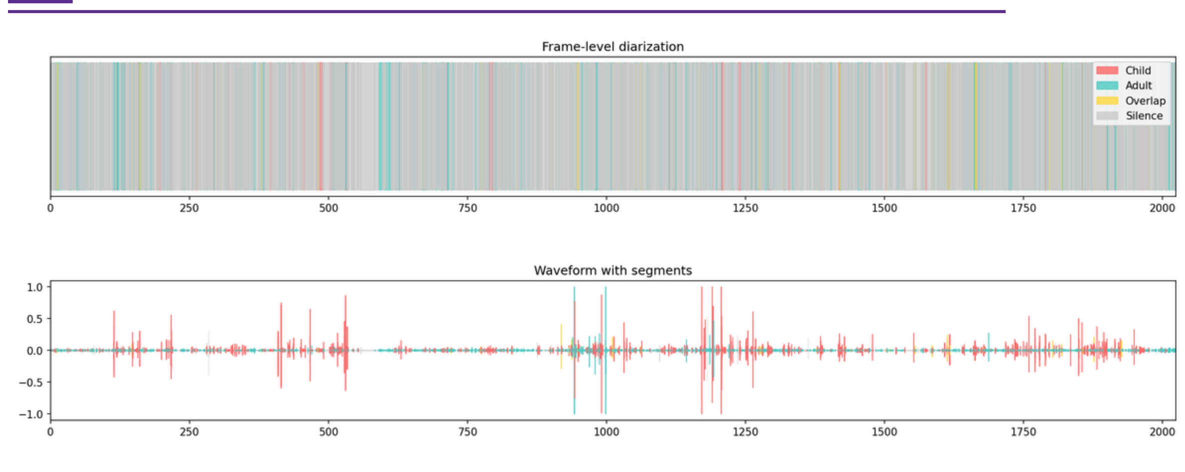


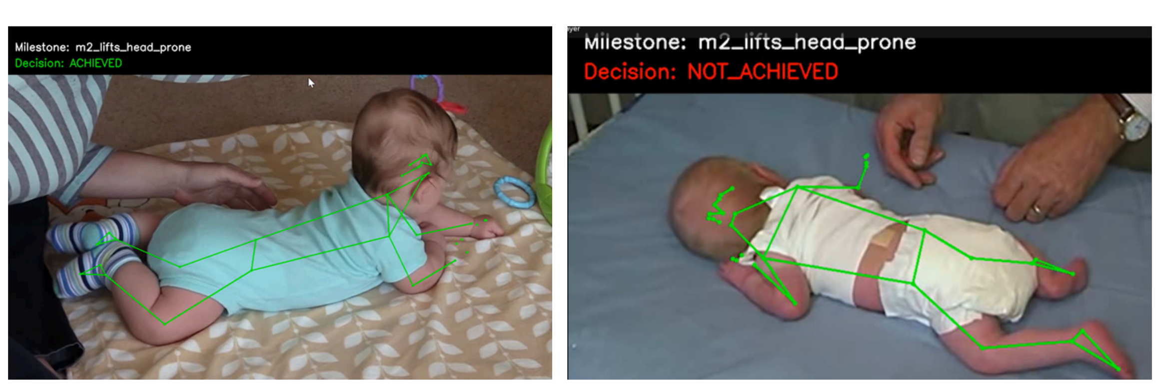
Database



Flutter

5 Results





Scan to view the full poster

6 acknowledgment

We would like to express our sincere gratitude to Liajlehum Association (جمعية لأجلهم) for their valuable support and collaboration. As an adopting partner of this project, their contribution in providing data and domain insights was essential in shaping, developing, and validating our system.

7 Conclusion

HAYATY addresses a critical gap in Saudi Arabia's child health ecosystem by transforming parental observations into structured, evidence-based developmental data through an Arabic AI-powered mobile app, enabling timely early intervention.

Future Work

→ Expand Arabic data collection